

Discrete Mathematics (Midsem Exam)

Mar 9, 2026, 9:30am to 12:00 noon

Total: 100 points

Answer all questions.

1. Give a self-contained proof that the set $\mathbb{R} \times \mathbb{R}$ has the same cardinality as $\mathbb{R}$. **10 marks**
2. Let F denote the set of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers. Let F_c denote the set of all *continuous* functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Is $|F| = |F_c|$? Justify your answer with a proof. **10 marks**
3. Suppose A is a subset of real numbers such that $A \times A$ can be decomposed as the union $B \cup C$ where B intersects every vertical line $x = x_0$ in at most finitely many points, and C intersects every horizontal line $y = y_0$ in at most finitely many points. Prove that A must be countable. **10 marks**
4. Derive the exponential generating function (egf) for putting distinct objects in k distinct boxes such that each box will have at least m objects in it. For $m = 1$ what is the egf and what does it count? **10 marks**
5. Find the exponential generating function for making teams out of n distinct players with a captain chosen for each team from its team members. **10 marks**
6. Triangulation of a convex polygon is putting nonintersecting diagonals between vertices of the polygon so that each region free of diagonals is a triangle. Let t_n denote the number of ways of triangulating a convex polygon with $n + 2$ vertices. Write out a recurrence relation for t_n and solve it to obtain a closed form for t_n using generating functions. **10 marks**
7. Let $(P, \leq)$ be a finite poset with more than mn elements. Show that P either has a chain of length $m + 1$ or an antichain of length $n + 1$. **10 marks**
8. Suppose $x_1, x_2, \dots, x_{mn+1}$ is a sequence of real numbers. Show that either there is an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$. **7 marks**
9. Let a_n denote the number of ways of writing n as a sum of distinct positive integers and b_n denote the number of ways of writing n as a sum of positive odd integers. Show *bijectively* that $a_n = b_n$ for all n . **8 marks**
10. Let $(P_n, \subseteq)$ be the subset poset where P_n is the power set of $[n]$. Let $B_n = \{0, 1\}^n$ be the set of all binary strings partially ordered by the relation $(a_1, a_2, \dots, a_n) \leq (b_1, b_2, \dots, b_n)$ iff $a_i \leq b_i$ for all i . Show that the posets $(P_n, \subseteq)$ and $(B_n, \leq)$ are *isomorphic*. That is, there is a bijection from P_n to B_n that preserves the partial order. Prove that $(B_n, \leq)$ can be decomposed as a direct product of n posets, each with 2 elements in it. Using this, prove that the Möbius function of P is $\mu(I, J) = (-1)^{|J \setminus I|}$ for all $I \subseteq J$ and $\mu(I, J) = 0$ otherwise. **15 marks**